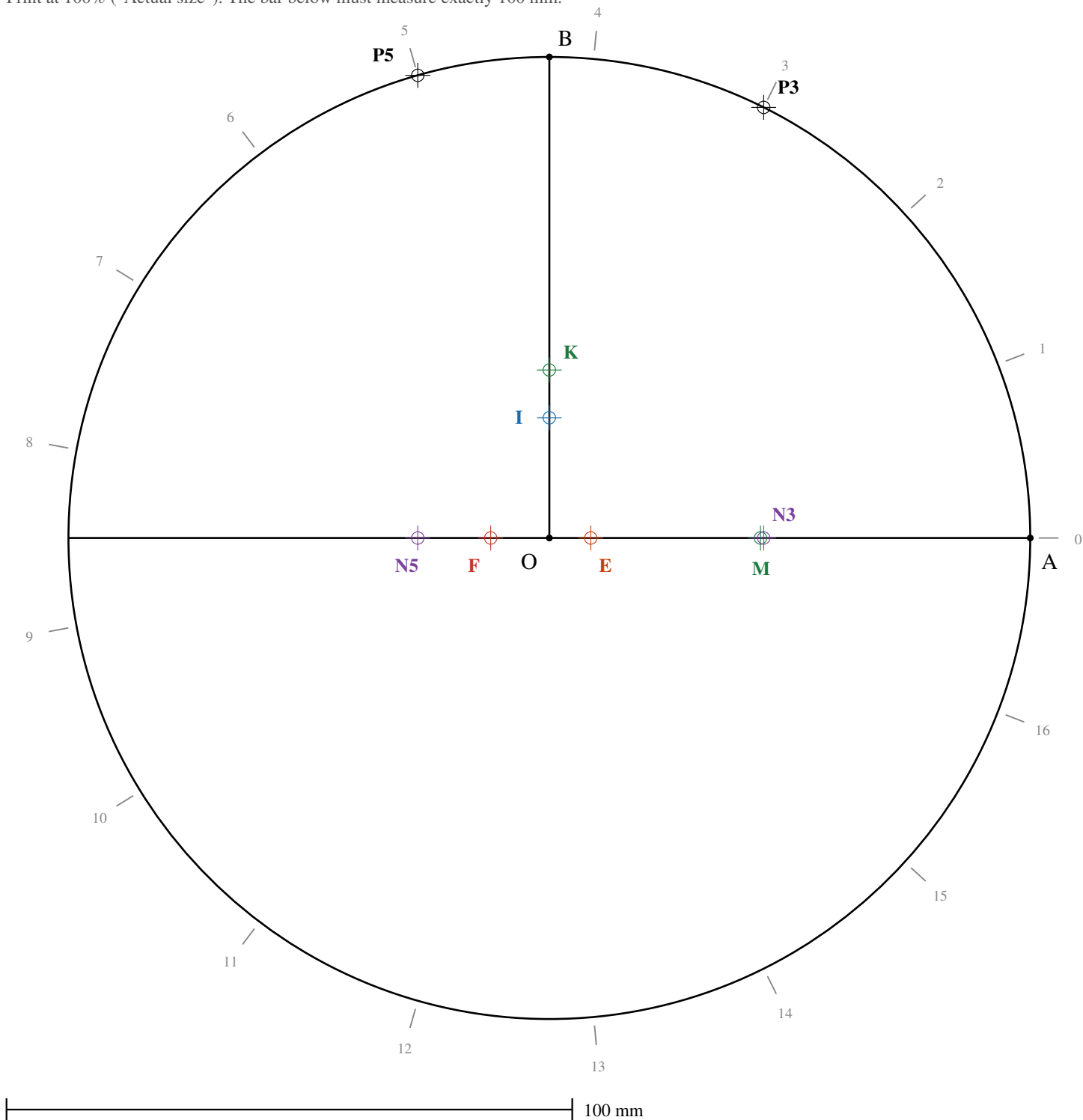


Gauss's 17-gon: Richmond's recipe, with the exact points

Print at 100% ("Actual size"). The bar below must measure exactly 100 mm.



1. I is a quarter of the way from O to B. Bisect angle OIA, then bisect that half again; it hits line OA at E.
2. At I, build a right angle to IE and bisect it: 45 degrees. That line hits line OA (past O) at F.
3. Arcs from A and from F (same width) give the middle of AF: M. Circle centered M through A meets OB at K.
4. Circle centered E through K meets line OA at N3 and N5. Raise perpendiculars there to the circle: P3, P5.
5. P3 is $\frac{3}{17}$ of the way round from A, P5 is $\frac{5}{17}$. Copy chord P3P5 from A to reach vertex 2; then vertex 2 to P3 is one side. Walk that width round the circle. The grey ticks show where all 17 vertices belong.

Note: M and N3 really are only about 0.6 mm apart. That is not a printing error.